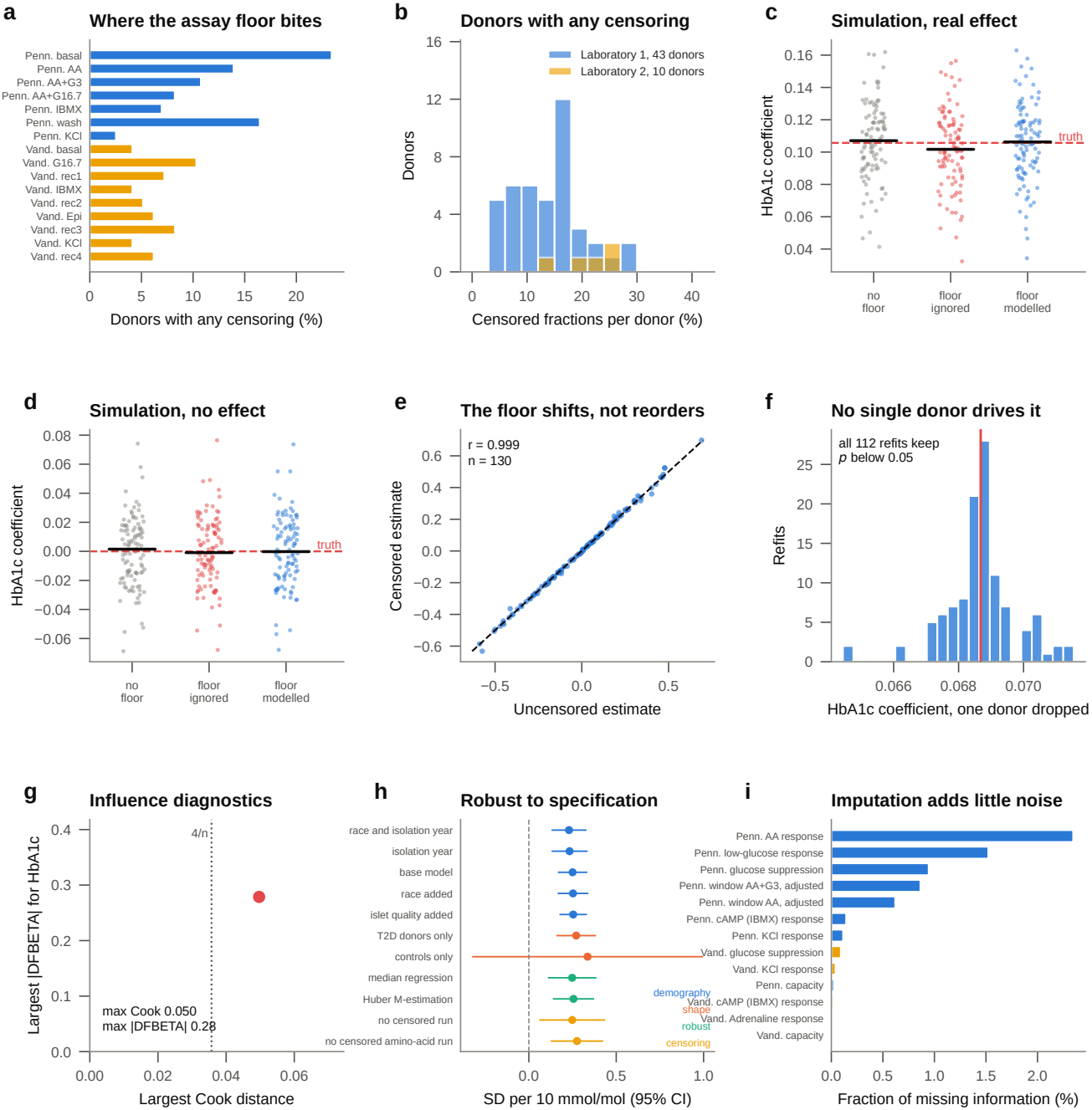
a
Where the assay floor bites
Penn. basal
Penn. AA
Penn. AA+G3
Penn. AA+G16.7
Penn. IBMX
Penn. wash
Penn. KCl
Vand. basal
Vand. G16.7
Vand. rec1
Vand. IBMX
Vand. rec2
Vand. Epi
Vand. rec3
Vand. KCl
Vand. rec4
0
5
10
15
20
Donors with any censoring (%)
b
Donors with any censoring
Laboratory 1, 43 donors
Laboratory 2, 10 donors
Donors
0
4
8
12
16
0
10
20
30
40
Censored fractions per donor (%)
c
Simulation, real effect
HbA1c coefficient
0.04
0.06
0.08
0.10
0.12
0.14
0.16
truth
no floor
floor ignored
floor modelled
d
Simulation, no effect
HbA1c coefficient
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
−0.06
truth
no floor
floor ignored
floor modelled
e
The floor shifts, not reorders
r = 0.999
n = 130
Censored estimate
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.5
0.0
0.5
Uncensored estimate
f
No single donor drives it
all 112 refits keep
p below 0.05
Refits
0
5
10
15
20
25
0.066
0.068
0.070
HbA1c coefficient, one donor dropped
g
Influence diagnostics
Largest |DFBETA| for HbA1c
4/n
max Cook 0.050
max |DFBETA| 0.28
0.0
0.1
0.2
0.3
0.4
0.00
0.02
0.04
0.06
Largest Cook distance
h
Robust to specification
race and isolation year
isolation year
base model
race added
islet quality added
T2D donors only
controls only
median regression
Huber M-estimation
no censored run
no censored amino-acid run
demography
shape
robust
censoring
0.0
0.5
1.0
SD per 10 mmol/mol (95% CI)
i
Imputation adds little noise
Penn. AA response
Penn. low-glucose response
Penn. glucose suppression
Penn. window AA+G3, adjusted
Penn. window AA, adjusted
Penn. cAMP (IBMX) response
Penn. KCl response
Vand. glucose suppression
Vand. KCl response
Penn. capacity
Vand. cAMP (IBMX) response
Vand. Adrenaline response
Vand. capacity
0.0
0.5
1.0
1.5
2.0
Fraction of missing information (%)